CSE 400

# HANDOVER PROBABILITY ANALYSIS IN DRONE CELLULAR NETWORKS

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# PROJECT OBJECTIVE AND SCOPE

Derive probability distributions for random variables and compute P[H(t)] analytically for SSM and DSM models.

- Understanding PDF/CDF formulas for all random variables
- Deriving nearest-neighbor distance distribution from PPP
- Generating simulation plots to validate theoretical distributions
- Explaining why DSM yields lower handover probability than SSM

# PROJECT SYSTEM OVERVIEW

**Mathematical Framework Flow:**

**1. PPP Theory**

Void probability formula → Nearest-neighbor distribution

**2. Distribution Assignment**

Assign PDF/CDF to positions, directions, speeds

**3. Temporal Evolution**

$x(t) = x_0 + vt[\cos\theta, \sin\theta]$

**4. Integration**

Condition on $(r, \theta)$, integrate over all values

**5. Handover Probability**

$P[H(t)]$ from geometric arguments and simulation validation

# SOURCES OF UNCERTAINTY

01 **Spatial Randomness**

PPP with density $\lambda_0$ gives void probability: P[no drone in A] = $\exp(-\lambda_0|A|)$

02 **Directional Randomness**

$\theta \sim U[0, 2\pi)$ creates isotropic movement patterns

03 **Speed Randomness (DSM)**

V ~ Rayleigh or Uniform introduces heterogeneous velocities

04 **Geometric Randomness**

Minimum distance $u^*(t)$ evolves unpredictably due to combined uncertainties

# KEY RANDOM VARIABLES WITH DISTRIBUTIONS

- **$\Phi D(0) \sim PPP(\lambda_0)$**
  Nearest neighbor: $f(r) = 2\pi\lambda_0 r \exp(-\pi\lambda_0 r^2)$ | $F(r) = 1 - \exp(-\pi\lambda_0 r^2)$

  **$\theta \sim U[0, 2\pi)$**
  $f(\theta) = 1/(2\pi)$ | $F(\theta) = \theta/(2\pi)$

- **$V \sim$ Rayleigh (DSM)**
  $f(v) = (v/\sigma^2)\exp(-v^2/2\sigma^2)$ | $F(v) = 1 - \exp(-v^2/2\sigma^2)$

  **u(t)***
  Distance to serving drone - time-dependent, computable for SSM

# PROBABILISTIC MODELS AND ASSUMPTIONS

01

**PPP for Positions**

- Tractable void probability formula
- Displacement theorem preserves PPP structure
- Models random deployment without clustering

02

**Rayleigh for Speed**

- Most drones near average speed
- Extreme speeds are rare
- Mathematically convenient

03

**Uniform for Direction**

- No preferred direction
- Ensures isotropy

# PROBABILISTIC REASONING

**Layer 1** - Spatial Statistics

PPP void probability → $f(r) = 2\pi\lambda_0 r \exp(-\pi\lambda_0 r^2)$

**Layer 2** - Temporal Evolution

$x(t) = x_0 + vt[\cos\theta, \sin\theta]$

**Layer 3** - Event Probability

Handover if new drone enters closer region

**Layer 4** - Integration

Condition on $(r, \theta)$, integrate with proper densities

# OPERATIONALIZING - SIMULATION & VALIDATION

- **Monte Carlo Simulation:**
- **Step 1**: Generate N ~ Poisson($\lambda_0$ × area)
- **Step 2**: Assign $\theta_i \sim U[0,2\pi)$, $v_i \sim fV(v)$
- **Step 3**: Update $x_i(t) = x_i(0) + v_i t[\cos\theta_i, \sin\theta_i]$
- **Step 4**: Track handover events
- **Step 5**: P[H(t)] = (handover count) / (total trials)
- Parameters: $\lambda_0$ = 1 DBS/km², v = 45 km/h, t = 0-100s

# CURRENT UNDERSTANDING GAPS

**Mathematical Details:**

- Exact derivation of Lemma 3 density formula
- Numerical evaluation of Theorem 2 integrals
- Full proof of Lemma 2 (once-leaving property)

**Conceptual Questions:**

- Why exactly does DSM < SSM intuitively?
- How tight is the DSM lower bound quantitatively?
- Sensitivity to distribution choice (Rayleigh vs Uniform)?

**Extensions Needed:**

- Different drone heights
- Random waypoint mobility
- Mobile users

# PLANNED REFINEMENTS FOR M3

**01 - Simulation Deep Dive**

- Understand Monte Carlo convergence
- Study confidence intervals
- Validate PDF/CDF plots match theory

**02 - Mathematical Validation**

- Verify all distribution derivations
- Check formula interpretations
- Ensure parameter choices are justified
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**03 - Team Coordination**

Everyone: Study simulation structure
Scribe: Document derivations
Reviewer: Verify analytical vs simulation match
GitHub: Organize plots and code

# SUMMARY OF M2 UNDERSTANDING

- M1: "PPP means random positions"
  M2: PPP has void probability $\exp(-\lambda_0|A|)$ giving $f(r) = 2\pi\lambda_0 r \exp(-\pi\lambda_0 r^2)$
- M1: "Theorem 1 equivalence is clever"
  M2: Works via PPP translation invariance - relative motion is what matters
- M1: "DSM can't get exact answer"
  M2: Different speeds break similarity transformation; Jensen's inequality explains direction

# THANK **YOU**